Department of Computing
Final-Year Undergraduate Project Scheme
2013-2014

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The department reserves the right to vary details of this project scheme. Any changes or clarification will be announced via Moodle.

[1] With thanks to co-authors of the previous versions: Neil Maiden, Jacob Howe, Julia Galliers, Andrea Zisman, Kevin Jones, Chris Smart and Tillman Weyde.

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1 Introduction

This document is the definitive handbook for students who are undertaking their final year undergraduate project in the Department of Computing. It defines the requirements, rules and guidelines that are specific to the 3-module final-year project. Other sections of your Course Handbook detail how your project mark affects your degree.

The undergraduate project is a module like other modules in the course scheme. As such, it has specified learning outcomes against which you will be tested throughout the project. These learning outcomes are listed in the Module specification, which you find on Moodle.

Responsibility for project scheme administration is delegated to the Projects Tutors (see table below). Administration will be dealt with by the Course Officer, Jagdeep Dosanjh (Jagdeep.Dosanjh.1@city.ac.uk) in the Programmes Office.

Course	Project Tutor	E-mail Address	Office
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2 The Essentials

This section provides the most essential information about this academic year's undergraduate scheme. It describes the project process, marking scheme and deliverables, at a glance.

2.1 The Process

The project process has been designed to meet staff, student and learning requirements, and is based on both positive and negative experiences with previous versions of the scheme in past years. Table 1 defines the key stages in the undergraduate project from the student's perspective. A detailed calendar of deadlines is on Moodle, IN3007 module.

Date/deadline	**Activities**	**Submission**
Summer before the final year	Student explores, negotiates and agrees project titles with candidate supervisors	None by student. Supervisor registers agreement through online tool
Start of Autumn term	For students without a supervisor and project: supervisor-project fair, bringing together students, supervisors and project topics at a one-stop shop for project selection, followed by a few more days for agreeing projects. (see details of process and allocation tool on Moodle)	
October	Enforced "auto-allocation" of unallocated students to supervisors according to availability, so that every student is able to start a project	
Student-supervisor meetings take place throughout the course of the project. At the end of every month, the students posts brief minutes of the meetings that have taken place that month: updates on the project progress and actions decided.		Monthly minutes of meetings, via Moodle
January	Supervisor and student meet for the **Project Progress Review** and student is given mark.	Review report
April	Students submit *Draft* project report for feedback from supervisor	Draft sent directly to the supervisor (no formal submission)
April	**Final Project Review**: review of project outputs by the supervisor and second marker; feedback from supervisor towards improving the final **Project Report**	Oral presentation
early May	Student submits final **Project Report** document	Project Report

Table 1 Undergraduate project process

The two official project submissions – the **Project Definition Document** (PDD) and the **Project Report Document**) indicated in Table 1 must be made to the Department of Computing offices and are treated as assessed coursework with respect to submissions and deadlines. They **must also** be submitted on Moodle (there will be detailed instructions before the deadlines). This method allows better feedback from supervisors and supports checks against academic misconduct. All submissions will be time-stamped by the Office staff.

The **Project Definition Document** will be returned to you by your project supervisor once it has been marked.

Please note that students who fail in their **Project Definition Document** or **Project Progress Review** or **Project Review** may be summoned to vivas by the project tutors: attendance to these vivas is **mandatory**.

To do yourself justice in both project work and taught modules, you must spend lots of time on each of them.

It is important to realise the scale of the project and time required to complete it. It is equivalent to 3 taught modules: it is worth 45 credits and is expected to take 450 hours of work. Given that the UG project lasts for 22 term weeks, you are therefore expected to spend on average 21 hours of work a week on it. Bear this in mind. Coursework and exam deadlines are no reason for you not to input the right amount of effort into the project.

Normally, your effort will vary throughout the period. It is especially important that you put in more than the average hours per week **at the beginning of the first semester**, so that you make serious progress before coursework deadlines get close. Your meetings with the supervisors will help ensure that this happens. Missing such meetings or not submitting reports on the meetings will lose marks at the **Project Progress Review.**

If you sit any exams in January, effort will be reduced over the Christmas period for exam revision, then halt during exam days in January. Immediately after the exams are over, project effort will step up to its highest levels, peaking at the submission date for the **Project Report Document**.

An important assumption behind this advice is that if you are registered as a full-time student you are indeed **studying full-time** for a degree. Whilst we expect that student may do some part-time work during the project period, work that is full-time or close to it is strongly discouraged. Requests for extensions due to this workload will be rejected.

Where to get advice:
about the rules for projects: • read and understand this document • follow the announcements distributed through Moodle, module IN3007 • if doubts remain, ask you supervisor and/or the project tutors: their Email addresses are listed in the Introduction and on line about how to manage your process for success, doubts about appropriate methods, report-writing: • advice is in this document
• Christian W. Dawson, The Essence of Computing Projects - A Student's Guide, London: Prentice Hall, 2000, and Christian W. Dawson, Projects on computing and information systems: a student's guide, Addison-Wesley, 2009. 2nd ed (in the library) and online: http://bit.ly/xTBXVL • for specific advice: ask your supervisor

Getting advice. Throughout your project, frequently *seek* and *note* advice, especially from your supervisor(s). This helps you to achieve *more*, achieve it *well*, and demonstrate that it *is your own* achievement, each of which earns marks.

Keeping your supervisor up to date about your work ensures that they see when to give advice, helps you to monitor what you are doing against your plan, and revise your plans if necessary, and is assurance against any suspicions of misconduct, which may instead arise if your supervisor has too little evidence of your work on your project. For all these reasons, it is also a good idea to keep a diary or blog of your progress (for instance, a short entry each week to note important activities, whether you are on schedule, any important news or changes in plans), which you can share with your supervisor and show at the end of the project.

Obeying the rules: It is important that you obey the rules set out in this document and the announcement posted on Moodle by the project tutors. Obeying the rules is easy, but violating the rules may cost you dearly. So, **read** this document throughout, and especially **check** the part concerning each deliverable while you prepare that deliverable. *If* you have doubts about what the rules mean for you, **ask** the project tutor or your supervisor! Never rely on rumours among other students.

Keeping records: you will be required to show (and submit with your final project report) your records of all phases of work, for instance notes from interviews and other meetings, raw data collected, and so on. So, in addition to or as part of your project diary, take notes and keep them, keep correspondence and Email that you exchange during the project, keep all kinds raw data collected at any stage. Preserve carefully both paper and electronic documents. See also the instructions for the Final project review and the appendices to the Project Report Document.

Writing up: Project marks come mainly from what you document in your **Project Report Document**, usually the hardest part of the work! Write up as much as you can during the course of the project, instead of leaving it all to the end.

Outputs: <u>All projects must deliver something of some value to identifiable beneficiaries</u> who have some relationship with the student such that the student can identify their requirements. These outputs are not your **Project Report Document**, but the main useful results that your project develops. Your **Project Report Document** must clearly indicate the output or outputs that the project produced, and their types. For a project to be acceptable, it must produce outputs of one or more types typically **from the following list**:

- A software prototype to be used by someone other than the student;
- A working software system (e.g. for a client);
- A specification and/or design of a system for a client;
- An evaluation of a working software or computer system, for a client;
- A software development methodology or component of a methodology (e.g. a process or technique), for use by a client or to be developed further by researchers;
- A detailed scientific result (e.g. from an experiment or study) to be used by a named academic researcher or a client.

If your planned outputs do not clearly fit in the above list, or for any other reason you are not sure that your project idea is acceptable, ask the project tutors. However, be aware that the results of *general surveys, literature reviews, business guidelines* and *discussion documents* are ***not valid project outcomes***. Any project that produces mostly these invalid outputs ***will fail***.

Important: Membership of the British Computer Society (MBCS) is an important professional qualification that can lead to Chartered IT Professional (CITP) Qualification. Our BSc Honours Degrees exempt you from parts of the BCS membership exams “on condition that students pass ***at the first attempt*** a practical problem-solving project.” So, miss the submission deadline, and you will not receive BCS exam exemption.

One comment on the *project fair*. During week 1 of term, the Project Tutors will hold a project fair, which will give students an opportunity to meet with potential supervisors. The supervisors will present their interests and the topics that they are prepared to supervise. Students will present their project ideas to relevant supervisors. Out of this fair we hope to marry up as many students to supervisors as we can. If you already have registered a supervisor and project title, there is no need to attend the fair and you should attend the start of the lecture and the other scheduled lectures for IN3007, which give important advice (these are early in the first term and in the Winter/Spring term).

There is a strict deadline for students to choose and register a project. After that, any remaining students without an allocated supervisor will be randomly allocated to available supervisors. This allocation will be rigorously enforced – there will be no scope for future

changes. The reason for this allocation is simple – to ensure that all students have a supervisor with whom to collaborate to produce a **Project Definition Document** by the deadline. All students will be informed of the allocation via e-mail.

2.2 The Project Marking Scheme

The project marking scheme is shown in Table 2. See your Course Handbook for the percentage of the final-year marks awarded to the final-year project.

Submission	Marks awarded (%)
Project Definition Document	5
Project Progress Review	10
Final Project Review	Must pass to proceed to submission
Remainder of marks after submission of Project Report Document	85

Table 2: Undergraduate project marking scheme

Late hand-ins at each stage will be penalised, just as lateness with any other coursework. Being late in any of the non-final deliverables will mean 0 marks, as for any other coursework. The standard rules apply for extenuating circumstances. If you are affected by extenuating circumstances and wish to apply for an extension, you must apply to the Extenuating Circumstances Panel. Consider that any such circumstances must be documented with proper evidence; also, consider that you are required to take proper precautions for events like brief illnesses (to be expected over the several months of your project), unless they occur right at the deadline, or computer crash, loss or theft of computers or storage media (for all such occurrences you must take precautions - like backups held in separate places - to minimise their impact on your work).

For the **Project Report Document**, in view of its great importance for your marks, two special rules apply:

- if you are late in submitting the Project Report Document , you will lose 5% of your awarded mark for that deliverable for each day or part of day that the deliverable is overdue. For example, if it receives a mark of 70%, but it is submitted one day and a half late, then the penalty applied will be twice 5%, that is 10/100 * 70 = 7%. As a result the final mark awarded to the **Project Report Document** will be 70-7 = 63%;
- if you fail to submit the Project Report Document, you will receive a module mark of 0.

2.3 The Project Deliverables

These are the project deliverables with their characteristics and requirements.

Deliverable	Characteristics and Requirements
Project Definition Document	A specification and rationale for the project. It describes what the process is meant to achieve, so that the actual achievements of the finished project can be assessed against it; and describes the work plan for conducting the project and monitoring progress
Progress meeting minutes	A brief text document containing minutes and actions of the progress meetings held with the supervisor
Project Progress Review	A presentation that will report progress of the project against objectives and plans, indicate any updates to plans, demonstrate that required project work has been undertaken to the level of quality required. The professionalism of the management of the project to this point will also be assessed.
Final Project Review	A presentation to the supervisor and second marker, and review by them, of the outputs of the project, including demonstration of software developed, when applicable. This is an **essential** step in the project: a project report **will not be considered** without it. Failing the review implies failing the project module.
Project Report Document	A report of the complete project that will enable the markers to assess the quality of the project

Table 3: Undergraduate project deliverables

The remainder of this Handbook describes these processes and deliverables in more detail.

3 Setting Up Your Project

One of the most important activities is to set up your project and produce a successful **Project Definition Document**. You will work on your project for over 7 months, so you will have to be interested by the project work and motivated to complete it. Your **Project Definition Document** sets out both your plan for your project and the criteria by which we will judge the achievements of the project, so it is an important deliverable.

3.1 Your Responsibilities at the start of your project

You are responsible for ensuring that you satisfy all requirements of the undergraduate project scheme.

You must also ensure you keep track of announcements to all students in the module, which are made via Moodle, and direct Email to you about your own project. In particular, all normal communication via email with your supervisor, project tutor, or the Programme Office should be via your University's email address. Check your University Email frequently and do not expect that Email can be sent to another personal email address that you may have (e.g., yahoo, google).

You must also check that you are enrolled in the IN3007 Moodle module and that you can use all its facilities. Report any problems with your Moodle or Email access promptly.

You also have responsibilities during the process of signing up to supervisors and/or their project titles. You might initially talk with more than one possible supervisor, until you reach an agreement with one of them; but it is important to conclude agreements quickly, and keeping two or more project negotiations open is irresponsible: you damage other students, and also yourself since the supervisor you wanted may be required to give the supervising "slot" to another student who is faster at deciding.

3.2 Registering Your Project Title and Supervisor

Project choices (student's details, the title of their project, and the supervisor) are recorded on an online tool.

After reaching agreement with you, your supervisor will register your name and the agreed project title. You will be able to check the record on https://www.soi.city.ac.uk/schoolintranet/ugprojects/paat/ that you are properly registered. This record is a written proof of agreement, which is binding for both.

The same online tool is also used for the late fast allocation scheme for those who have not found a supervisor by early October. We will announce this phase of the process on Moodle.

3.3 Academic Supervisors and their Responsibilities

Academic supervisors are experienced in project work. In choosing your supervisor, take care that their expertise and availability are compatible with your project needs.

Your academic supervisor's responsibilities towards you include:

- Helping you, before agreeing to supervise you on a specific project, to ensure that your project objective is clear and focused enough, and that it is feasible for you to achieve it and demonstrate that you did;
- Helping you ensure that your project work plan is complete, appropriate to your project objective, and will enable you to demonstrate understanding of the relevant state of the art;
- On request by you, providing advice on how to carry out your project work or referring you to a source that can provide such advice;

- Giving feedback on your work, including your write-up, when sought in good time by you, appropriate to the amount of time you have available before the relevant deadline(s);
- Providing assessments of your deliverables suitable for feedback to you.

If you have concerns over these responsibilities, contact the relevant Projects Tutor. The Supervisor's Charter is included as an appendix in this document (Appendix B: The Supervisor Charter).

3.4 Some Simple Rules

We have developed some simple rules that we will apply very strictly during the project selection and definition processes. Please read these rules carefully and ensure that your project obeys them all. If you have doubts, please contact your Project Tutor. We have given each rule a simple name for you to be able to refer to it during the project definition phase and to communicate your questions to the Project Tutors.

3.4.1 The Definition-First Rule

No other project deliverables are valid until you have submitted your **Project Definition Document** in accordance with *all* of the 6 rules specified in this section.

3.4.2 The Sol-Supervisor Rule

Your academic supervisor must be a member of the academic staff of the School of Informatics, unless you get special approval from your Project Tutor. If you wish to be supervised by someone who is not an academic in the School, contact the Project Tutor as soon as possible for advice.

3.4.3 The Supervisor Approval Rule

Your choice of project topic is only valid if approved by your supervisor.

3.4.4 The Business Computing Rule

For BSc Business Computing Systems students, the project must be about some aspect of computing in business. However, the focus remains on ***computing*** in business, rather than business per se. If you have concerns that your project will be too business focused for a Business Computing title, talk to the relevant Project Tutor.

3.4.5 The Non-Build Rule

For students who are **not** Business Computing or Information Systems students, the project must have a substantial design-and-build component. Topics with no scope for software construction require the Course Director's written approval; if such approval is given, your **Project Definition Document** must explain the project's validity for your course.

3.4.6 The Unique-Project Rule

Ordinarily, no two students can have projects with the same objectives. Any exceptions to this rule require explicit and specific approval by the relevant Project Tutor at the outset and must be explained in each **Project Definition Document**. The list of titles already chosen will be visible on line, so you can check with your supervisor and avoid duplication.

3.5 Choosing a Project Topic

Many students are initially uncertain about a topic, so we summarise here some guiding principles.

The common feature of all good project topics is that they (i) address issues in Informatics and (ii) allow a student to tackle challenges and demonstrate professional skills.

The options are: (i) to build software or (ii) to answer some (challenging) questions (usually by experiment and observation); or - very often - a combination of the two activities. "Answering some challenging questions" is called "research" if the questions are important for scientific research (your supervisor will help you to define research questions that are challenging enough and yet can be addressed within the duration of a student project), or they may be questions that are important for a client (and then it is the client who specifies the question).

Many supervisors offer project topics for students to choose from. Choosing one of these topics, or a topic proposed by an outside client, has the advantage that you will have to solve a concrete problem for someone. So, it adds motivation, challenge (you need to understand what is really needed from you, and interact with the person you want to help, so that you deliver a real solution) and help (there is interaction with an expert in the problem domain).

Some students postpone a choice too long, looking for topics that seem “interesting”. This is a mistake. You want to have chosen at least one week before our “auto-allocation” deadline. If you are interested in pursuing a specific topic, by all means start early to look for an appropriate supervisor. But in any case, do not delay too long: the sooner you start your project, the more time you have to make a good start and in the long run a good project. Consider seriously the topics that supervisors offer, and if in doubt, just choose one! Once you look into one of these problems in depth, then the commitment to solve it will bring passion for the project, not just "interest".

3.6 Project components and types

Every project belongs to **one or more** of these 3 types:

- Design and build projects;
- Client-based projects;
- Academic research projects.

Some student projects will fit neatly into only one of these 3 types; but many projects will belong to two types or all three types. For each type there are specific prescriptions about what to include in your PDD and planning. So, for instance, if you do a client-based project that is a design-and-build project as well, read in this document the parts concerning both types, and follow both all instructions specified for client-based projects and all instructions specified for design-and-build projects.

3.6.1 Design and Build Projects

Design and build projects are available to all students. These types of projects involve design and construction of a software application or tool intended for users other than the student. The design and/or construction must be non-trivial. The development should follow an established software engineering method, and any enhancements to or deviations from the planned method must be discussed and justified in the **Project Report Document**. In exceptional cases, we will permit projects that involve analysis and design but not a build, with an intention that the design can be implemented by a third party.

In planning Design-and-Build projects and producing the **Project Definition Document** you should state:

- The purpose of the software;

- In what way the project is novel;
- What theory (if any) underpins the project;
- Applicable software engineering methods;
- What tools will be used, so far as decided;
- Methods envisaged for testing and evaluating the software;
- How the complexity of the work merits it being a final-year project.

In a design and build project, you are entitled to reuse existing software, provided you have a right to use it (e.g., software that is publicly available or software provided by your client) and that this reuse is clearly and completely acknowledged at all stages of your project. Re-use is good software engineering practice. However, it is unacceptable to rely solely or extensively on such software to implement a Design and Build-type project. Such projects require students to design and develop a software system on their own: they must include an amount of development work by the student comparable to that required for projects in which no re-use takes place. Furthermore, all reuse of pre-existing software should be agreed with the supervisor beforehand. If you have any doubts on this aspect, discuss them with your supervisor and if necessary with your project tutor.

3.6.2 Client-based Projects

Client-based projects are available to Business Computing and Information Systems students, and to other students if the project includes a significant design-and-build component. These are undertaken for a client, often external to the University. They need not involve software construction. They must produce outputs (e.g. evaluations, analyses, proposals) that are approved by the client. The project can involve work, methods or outputs that are novel in the client environment. Projects involving the design or specification of software but not its construction normally fall into this category.

In planning Client-based projects and producing the **Project Definition Document** you should state:

- The names and addresses of the client(s) and the name, telephone number and e-mail address of each contact person;
- The client problem to be addressed;
- The resources to be made available to the student (staff time, data, equipment etc.);
- The outputs required by the client;
- How the outputs may be reviewed;
- The intended use of the project outputs;
- The novelty of the project to the client(s);

Last, you are required to provide a summary page that states the client's detail and general conditions for the project, as described in an appendix of this document (Appendix A: Client information sheet, to be Submitted with Project Definition Document at Start of Project). The School will ask the client to confirm acceptance of the project and that the details are correct.

Note that at the end of the project the School will again contact the client and ask them to state the assistance given to you, how they have assessed your work and how satisfied they are with the outputs. You need to forewarn the client that the School will require information in these two occasions, and that their replies will be essential for the assessment of your project by the School.

3.6.3 Academic Research Projects

Academic research-only projects are available to Business Computing and Information Systems students only; Computer Science and Software Engineering students may choose academic research projects only if they are also “design-and-build” projects. Academic research projects are undertaken to investigate a research question specific to a **computing,** rather than to a **computing-related**, topic. If you have any doubts about the suitability of the research question under investigation, contact your relevant project tutor. Furthermore, an academic research project must contain a research contribution from the student that is equivalent in effort to, for example, the development of a model or the construction of a piece of software. Although research projects might include data gathering in the form of experiments, questionnaires, interviews and web-based forms, the gathering of data in itself will not constitute an acceptable level of research effort by the student. **Rather, the project must produce some rigorous analysis of the data and/or some other output, based on the data, that can be used for further research by a third party, for example the supervisor, an external body or other stakeholders in the project**. Furthermore, academic research projects must be designed so that it is clear what factors affect the validity of the results and to what extent these results can be generalised.

Research projects can bring great challenges and satisfaction, but students are often unclear about the requirements of a research project. Research at university level may be very different from what was called research in earlier schooling. **Broad and general questions make bad research projects**. Narrow questions, such that a student can produce concrete new knowledge, make good projects.

For instance, asking what effects a certain new technology will have on business is not an appropriate topic: you would not be able even to read all the material already produced by professional researchers to answer the question. But a careful, professionally run study of the effects of that new technology on a specific company, or a small experiment to check whether a conjectured effect on a certain type of users does occur, will often be a very good project, adding new knowledge to what is already known about the subject. So, take advice from your supervisor: check that you agree and understand the research questions to answer and that you know what is required for your project to succeed,

In planning Academic Research projects and producing the **Project Definition Document** you should state:

- The research question (or questions) to be addressed;
- Any research initiative or project (e.g., a research project run by your supervisor) that your own project is a part of;
- The research methods and tools to be used;
- How you will judge the validity and generality of your results;
- In what ways the project may contribute to related research activities.

3.6.4 Combining Project Types

Many student projects will not fit neatly in one of these 3 project types: for instance, you might build software for a client, and/or as part of a research project; a research project may have a client. It is important for you to recognise to which project types your project belongs. In specifying a project that has features of more than one type, your Project Definition Document must cover the union of the respective lists of details to be stated.

Business Computing and Information Systems students can define projects with characteristics from all 3 project types. In contrast, other students must define a Design and Build project, which may also belong to the other two types.

3.7 Thinking Ahead to the Project Report Document

It is important when setting up your project that you think ahead to the shape and content of your final project report. After all, you are stuck with it for the next 6 or 7 months. Think forward to how you will undertake the planned work, and complete the work well so that you will gain good marks for the work. It is important that you try and envision how the project result will look, even at this stage, as it will guide you throughout the project.

3.8 Your Intellectual Property Rights

The ownership of Intellectual Property (IP) that might arise from a student project is always a difficult subject. The University has established a policy, which you can check online. The UG project process adopts and implements this policy. Some relevant rules are described here.

The initial presumption is that where a student has developed IP that they wish to commercialise, the student owns the IP. However, the following exceptions might apply:

1. **Students who are sponsored**. A condition of sponsorship may be that the sponsor may own any IP developed during the period of sponsorship. Sponsored students are, therefore, advised to check the terms of their sponsorship agreement;
2. **Students working on a sponsored project as part of their project**. In this case, the sponsor may own any IP that they develop. This will be part of the research contract and the supervisor or department, prior to starting the project, should inform relevant students. In such cases, they will be required to assign any resultant IP to the University;
3. **Students who are working on a project that derives from the IP of academic staff** – typical in academic research-type projects. If such projects result in a commercial opportunity that the University wishes to exploit, the student will be required to assign their IP to the University.

As for copyright in the project report and other submitted deliverables and results of project work, and exception to your rights as author is that the University retains the right to copy them freely for assessment or other internal purposes, including submission to plagiarism-detection services (Turnitin). Copies of reports may be provided to other students as aids for their own projects or other parts of their studies, unless the author explicitly indicated that they contain confidential material not to be disclosed. In addition, if you pursue a client based project we may give the client copies, or show the client any material that you submit to us as part of your project.

4 The Project Definition Document (PDD)

The **Project Definition Document** is the specification or blueprint of your project. It is the basis on which the project is planned, conducted and assessed. You should start work on the **Project Definition Document** as soon as possible, if possible have a draft reviewed by your supervisor for comments, and **submit 2 copies** to the Department Office by the deadline published on Moodle, as well as submitting through Moodle. Each paper copy must have a signed coursework cover sheet. ***Make sure you attach all documents required: an incomplete submission is not considered as a valid submission***.

The most important features of the **Project Definition Document** are the specification of testable project objectives (what will you produce, how will you decide whether it satisfies the project's goals?), the research questions and/or research hypotheses, if any, and the work plan that will deliver the results.

The following subsections describe the structure we expect for your **Project Definition Document**.

Your **Project Definition Document** should be up to 1000 words in length, excluding the basic information given on the first page of the document, the references section, and the attachments (Ethics checklist and client information). Head it "**Project Definition Document"**. 1000 words are enough for the required content of the **Project Definition Document**. Penalties will be applied to submissions that ignore these submission rules.

Please note that the department reserves a right to reject or require changes to any **Project Definition Document** at any reasonable time, if needs or circumstances arise.

4.1 Basic Information

The first page of the **Project Definition Document** must include:

- The title of the degree you are studying for;
- A short, informative project title;
- Your name and contact details including phone number and email address;
- The names and full contact details for your Supervisor(s) and any external client(s);
- Full identification of who proposed the project (you, your client, your supervisor, or someone else?);
- A concise project description;
- A brief outline of all arrangements for proprietary interests in your project work and/or outside help. Attach in full all documentation of agreements. If there are no such arrangements, this must be clearly stated;
- Any promises you are making in order to secure acceptance of the project (e.g. by your supervisor or projects tutor);
- Your signature.

Furthermore, if you have clients, documentation of your agreement with them and of their acceptance of certain required conditions must be included (read Appendix A: Client information sheet, to be Submitted with Project Definition Document at Start of Project).

In some circumstances you will also be required to provide documentation of a sponsorship arrangement, in which you get some benefit from assigning ownership or a licence to use your intellectual property.

Subsequent pages of the **Project Definition Document** shall include the contents of the project definition, divided into ***6 main sections*** – problem to be solved, project objectives,

project beneficiaries, work plan, project risks, and references (at least one relevant piece of literature).

Last but not least,

- ***every*** student must fill in and attach at the end of the **Project Definition Document** a filled-in **Research Ethics Checklist** (see section 4.8, and the School of Informatics Research Ethics Policy document on Moodle and the Intranet), and
- those with client-based projects must attach a Client Information Sheet (see Appendix A: Client information sheet, to be Submitted with Project Definition Document at Start of Project).

PDDs without the Research Ethics Checklist, and PDDs for client based projects that lack a Client Information Sheet, get zero marks and the missing parts must be submitted before the project can proceed.

4.2 Problem to be Solved

This section shall include a clear and concise statement of the problem to be overcome or solved during the project by the student.

For Build and Design-type projects, the problem might be the lack of a software system to fulfil a purpose, a complex or difficult design, an implementation problem to overcome, or an unexplored opportunity that has arisen from advances in information technologies. For Client-Based-type projects, the problem might be a commercial problem that a client requires an IT solution for, a failure with a client's current IT system, or a requirement that a client has for some new IT solution. For Academic Research-type projects, the problem might be a research problem, identified by your supervisor, that requires a specific solution, an evaluation of new technologies, a standalone piece of research in a larger research project, or the development of a new process, method or techniques for developing software-based systems.

4.3 Project Beneficiaries

This section shall describe who will benefit from the project. For example, Design and Build-type projects should benefit future stakeholders such as owners, further developers or end-users of the software. For Client-Based projects, beneficiaries might be the client organisation, its stakeholders or its customers. The results of Academic Research-type projects should benefit members or parts of the scientific and technical community first (e.g. members of academic or research staff, PhD students, research projects or broader areas of academic endeavour), and hopefully end-users and/or other people later. Whatever the project type, try to specify the expected benefits in a quantitative or anyway clearly defined way, so that at the end of the project you will be able to demonstrate that you have met the project objectives.

Whoever the intended beneficiaries are, you need to plan your project to deliver results to them.

4.4 Project Objectives

A successful project will have well-defined objective(s) and, where relevant, research question(s) and hypothesis/es. All project definition must give an explicit project objective. The objective is what the project is aimed at achieving – it is what <u>you</u> aim to do and achieve during the project. It needs to be explicit and unambiguous to avoid different interpretation by different stakeholders. There should be a single main objective – this will give the project a clear focus. You should define the objective with a key sentence, with a subject and verb such as:

- “This project shall ...”

This will normally then be broken down into sub-objectives, that have to be met for the overall objective to be met. When defining these, think about how you will *test* whether they have been met at the end of the project. In your **Project Report Document** you will be required to demonstrate application of this test (or these tests) and report the result (objective is met, not met or partially met). Therefore it is important at this stage to identify simple tests that you will apply during the project, and to state this test for each objective or sub-objective.

Academic research projects, and some projects of other types, also require research questions and/or hypotheses. A research question is a question about the subject of your project that you set out to answer in your project. At the end of the project you should have data and results to enable you to answer each research question. The subject and verb of a sentence specifying a research question should describe the subject domain of interest, for example:

- “Will the XXX method improve ....?" or
- “By how much will the use of the YYY software tool reduce …?"

For some projects it will be appropriate to define hypotheses that your research will test. A hypothesis is a tentative proposition that is subject to verification or refutation through your subsequent investigation. Hypotheses propose potential answers to research questions.

It is important again to think about how you will discover answers to research questions or will test (typically by attempting to refute them) hypotheses that propose answers to them. As with project objectives, you need to identify these methods explicitly at the project definition stage and to describe their application and results in your **Project Report Document**.

4.5 Work Plan

You will need to undertake a systematic piece of work to achieve your project objectives and/or answer your research questions. To organise and subsequently describe this work, you need a work plan based on appropriate methods. Produce a simple work plan divided into work packages with their outputs, resources, start-dates and end-dates. Justify the work packages by reference to the objectives, research questions and hypotheses. You should also show that the work plan is feasible.

When you write your PDD, you may have already a detailed idea of your method – how you will accomplish your results – (for instance, “I will program this product in C++, using the libraries available from company so-and-so and test using the tool available from my client”), or parts of it may yet to be decided (for instance, you will need to find which libraries are available, or which programming language is most suitable). Be clear about these aspects and try to plan how and when you will finalise the parts of your method that are undecided. This will help both you to reflect on how best to plan your work and your supervisor to give you useful advice.

Where appropriate, include a simple diagram of the plan with a time axis (e.g. a Gantt chart). An activity diagram may sometimes be appropriate.

You should later use the work plan to control your project: keep track of your work and compare it against the plan. Deviations will indicate that you need to change something to ensure the success of the project. When reporting progress to your supervisor, you should refer to this plan so that he/she can advise about responding to deviations.

It is important for you to relate the planned work to proposed deliverables and outputs from the project. There are 2 types of documents and artefacts that projects must deliver, which we have distinguished by different names:

- the first type is the ***deliverables*** that you produce ***for the University*** as part of your UG project process, such as the **Project Definition Document** and **Project Report Document**. These are produced by all students at the required times;
- the second type is the ***project outputs*** specific to your project, and needed by your client, sponsor or the individuals or organisations who will be the immediate

beneficiaries of your project. Think carefully about what outputs these people will need (e.g., software, reports and documents of various types), and define the work plan so that it delivers them. Remember – in your final **Project Report Document**, you will be required to describe these outputs that you have produced, and marks will be awarded for them. So plan for them now – you know it makes sense.

4.6 Project Risks

The **Project Definition Document** (PDD) should demonstrate your understanding of the main risks that are associated with your project and your plans against them.

Project risks can arise in many forms. Examples include:

- Lack of resources, such as the absence of your supervisor, client or important computing resources;
- Changes in the requirements of the project client or sponsor at key stages in the project;
- Serious failures in the technology, including programming problems that cannot be resolved, hardware failures and incorrect assumptions you may have made about what can be achieved.

For the main risks - events that may quite plausibly happen and if they happen would cause serious damage to your project - you need to have an idea of how you can either make them less likely to happen, or a fall-back plan to limit damage if they do happen. In your PDD, you are required to identify these risks and these defensive strategies. This allows you to be prepared, and your supervisor to give you advice from their own experience.

You must anyway make provisions against those risks that are common to all projects (e.g., you should make frequent backups of your work, to be protected against disk failure or theft), ***but*** in the PDD it is important instead that you analyse those risks that are more specific to your own project, e.g. if it requires you to become familiar with an especially complex technology, or there is a concrete risk of a late change in your client's requirements, etc.; and that you outline your strategies and fall-back plans regarding them. This planning is very important for the success of your project.

4.7 Referencing One Important Piece of Work

All projects, no matter what the type, should be based on existing knowledge reported in one or more books or academic papers. In your **Project Definition Document** you are required to cite one such paper or book and describe its relevance to your project. If you are undertaking a Design and Build-type project you might want to cite a paper about the application domain, the software engineering method that you will follow, the implementation technologies that you will employ or the technological problems to overcome. If you are undertaking a Client-based project you might want to cite a paper about the method that you are following, the client's problem, or the candidate solutions that you will consider. If you are undertaking an academic research-type project, you might want to cite a paper about the relevant academic theories, models or approaches.

As with all your work, any material authored by others (e.g. your supervisor) that you include in this document must be correctly attributed. See the chapter about good professional practice, and check in particular the instructions about correct referencing and about plagiarism.

4.8 Research ethics issues

It is your responsibility to pay attention to ethical issues that may arise from your project. These concerns are ***not limited*** to research type projects. Projects should be planned and executed with adequate safeguards for the rights of all people, and projects must not proceed if these guarantees are not possible or are not provided.

A specific set of rules applies when a project (research or not) studies people or data about people. More precisely, quoting from the School of Informatics Research Ethics Policy document, all projects

> "involving human participation [...] should undergo ethical review. In REP [research ethics policy], 'human participation' covers direct data collection from people, for example surveys, observation and physiological measurement. It also includes retrieving data from individual people records. Research involving anonymised secondary data, such as reviews of published or otherwise disseminated documents and analysis of aggregated data, does not require ethics committee approval. Research involving animals requires ethical approval too, but it is not covered by REP and should be submitted for consideration by the Research Ethics Committee of the Senate

Every student must assess his/her project in view of ethical principles, and safeguard the rights of all involved. The reference document to use is the School of Informatics Research Ethics Policy, which you will find via the IN3007 module on Moodle. You must read this document to understand the possible ethical issues, and follow the instructions it contains.

You must consider ***early on*** the possibility of these issues arising in your project, and discuss them ***early on*** with your supervisor. Otherwise you risk planning a project that cannot be allowed to continue, or that will be delayed heavily. Consider that ethical issues may not be obvious when dealing with people's participation in the project and collection of data about them. Your supervisor will advise you, if you ask, according to his/her experience, and may obtain advice from the School's Research Ethics Panel for guidance about difficult questions.

You need to attach to your PDD the Ethics checklist form from the School of Informatics Research Ethics Policy, filled in by you with reference to your planned project. Follow the instructions also to decide whether you need to prepare a consent form for people taking part in the experiment, and if so, attach it to your Project Definition Document.

An important part of the feedback you receive to your PDD is the supervisor's advice as to whether there are unresolved ethical issues, and how to deal with them. This may include any of these options:

- applying specific safeguards for the rights of the people involved (e.g., processes for guaranteeing confidentiality)
- dropping some activities and redesigning the project accordingly so as to keep it viable and making it acceptable from the ethics viewpoint. In this case, you will need to submit a new version of the PDD, in which these problems are corrected. Your supervisor needs to approve the corrections before you can proceed with your project
- indicating that the project would need approval from the Senate Ethics Committee; in this case, you will need to discuss with your supervisor the implications about the timing of your project, and in rare cases the project may become infeasible
- indicating that the project cannot be pursued, because it raises unsurmountable ethical problems.

You see that it is essential for you to consider these issues before you submit your PDD.

4.9 Helping You to Write the Project Definition Document

The following information sources are available to help you with your choice of project and your **Project Definition Document**:

- https://www.soi.city.ac.uk/schoolintranet/ugprojects/ug-p3-prjs/ and linked pages which include:
  - Project topics proposed by academic staff;

- Staff project supervision interest areas.

- Your Project Tutor for advice on potential supervisors for given topics and on all aspects of the project scheme, especially on choosing your project;

Last but not least, as with your main report, you are strongly encouraged to seek your supervisor's comments on drafts of your Project Definition Document.

5 Second markers

The Project Tutors will allocate second markers to all projects after the **Project Definition Documents** have been submitted at the end of week 3 and before the time set for Final Project Reviews. The exact timing in each year will depend on the allocation of supervision on other courses in the School. The project tutors will inform students of this allocation.

Note that the Project Tutors are responsible for the allocation of a second marker to a student's project because it is an academic rather than an administrative matter. Students have no input to this process.

6 The Project Progress Review

The student shall meet with his or her supervisor to present the progress on the project to that point.

The **Project Progress Review** will test 3 student learning outcomes:

- To be able to manage (i.e. plan, monitor and control) a large individual piece of work to arrive at an effective outcome;
- To be able to undertake a large and individual piece of work without frequent supervision;
- To be able to design a systematic, suitable and repeatable working method that demonstrates appropriate and effective research skills for a large and individual piece of work.

The required content of the **Project Progress Review** reflects these learning outcomes and tests. You are required to deliver, at least briefly:

- an objective review of the work undertaken so far, with explicit evidence that you have monitored your progress and identified any deviations;
- an overview of your progress against the original work plan, considering the need to control and re-plan in light of the results so far. Most of you will need to refine or change their plans to some extent as a result of experience in these first two months: you will report on these changes;
- a clear definition of the actions you have undertaken to ensure that your project satisfies the work plan, and clear and reasoned changes to the work plan;
- a summary of key sources you have read (references for your project report) and their relevance to the project as it is progressing.

Your **Project Progress Review** will take place at a time agreed with your supervisor during the period indicated in Table 1 (the exact time window will be specified on Moodle). You are required to attend this meeting and to deliver a Project Progress Presentation during the meeting. Your audience should include your supervisor and fellow students. Failure to attend will result in a 0 mark for your **Project Progress Review**.

The supervisor will ask each student in turn to deliver the presentation.

Your presentation should last for between 5 and 10 minutes**.** It must be an oral presentation. You can produce hand-outs to hand out to your supervisor and fellow students during the session: e.g., diagrams illustrating your plan and work, or lists of references. Do not bring slides or PowerPoint presentations. At the end of the presentation both the supervisor and students can ask questions. This meeting provides an ideal opportunity to check your progress against that of your fellow students, to compare notes and to learn from each other.

After the meeting the supervisor will award a mark for your presentation and progress made so far. Supervisors will identify to the Project Tutors students who are failing at this stage. Your mark will also reflect your attendance and performance in Progress Meetings prior to this, and the professionalism of the meeting reports submitted.

7 The Draft Project Report Document

Your Draft Report is simply a draft of the **Project Report Document**, which you deliver to your supervisor (not to the office) to get his/her advice.

It must include a full draft table of contents (covering any sections that are yet to be written) and clear indication of how you plan to evaluate your work (to enable feedback on this).

Your supervisor will give you feedback on this draft, e.g. regarding major omissions, defects of clarity or relevance, inadequate arguments or unsupported claims.

This deliverable carries no marks, but the feedback you will receive is very important to ensure that you produce a good project report document. Most students are novice writers, as far as this kind of report is concerned, and may lose marks through naïveté` if they do not submit their draft for review.

Get it to your supervisor sooner, and you get more feedback. Miss the deadline, and there are no mark penalties, but you risk missing vital advice that you need. Supervisors have many drafts to read. A supervisor will not be expected to provide feedback on a whole or substantial part of a project report draft in less than 2 weeks of first receiving the report draft. This has serious implications for planning your project.

We advise you to talk in advance with your supervisor about what you will deliver for review, especially if you expect not to have the whole draft ready by the deadline. Supervisors can advise you about prioritising what you submit to them (and occasionally they can promise feedback on late drafts. But do not count on this: each supervisor has many students and other commitments).

8 The Final Project Review

Every student is required to demonstrate the outputs of their project to their supervisor and second marker **together** before the final submission (the exact dates will be on Moodle). This is a mandatory process, required so that your supervisor and second marker can directly see the outputs of your project (separately from their description by you in your project report) and ask questions about it. A typical review will last 20 to 30 minutes.

The review serves several purposes, of which the main ones are:

- communicating information to the markers about the project and its results, and introducing the second marker to the project;
- demonstrating your ability to explain your work, and understanding of your project method and outputs;
- protecting against academic misconduct by demonstrating that the student can explain the materials, methods and results of the project;
- giving you useful feedback, indirectly through the questions of the markers or through direct advice, about how to present the work in the project report document.

The review carries no marks, but unless it is performed successfully, the project will be deemed incomplete, and a mark for the **Project Report Document** will not be assigned.

There are only three ways for a review to fail:

- a student does not show up for it
- a student does not show all the material produced
- a student does not demonstrate knowledge of his or her own project.

Scheduling

The student must make himself available for the Final Project Review in the period specified in the project calendar in this document, and attend at the time set for the review. The exceptions are: (i) the review must not conflict with scheduled lectures or tutorials; (ii) a review may be scheduled during vacation time for common convenience between student and markers, but non-PP students may request that it take place outside vacation time instead. Times for the reviews will normally be set by the supervisors and communicated to the students by Email. The student is responsible for informing the supervisor if either of the two exceptions above arise, or if the student did not receive a notification of the time for the review. This last requirement is to guarantee that no student finds himself at the end of the project without a review, even if there were communication problems.

We anticipate that the Final Project Review will take place on University premises, although certain outputs of client-based projects might need to be demonstrated at the client's premises. In such cases you need to inform your supervisor well in advance.

Failing the Final Project Review implies failing the project. To prevent this, failing students will be summoned to a viva, within a week of the first review, to present any missing material and show their understanding of the material they present.

Contents

Clearly the different types of projects will produce different outputs. All types of outputs will need to be reviewed. For example:

- Design-and-build projects will deliver software products and prototypes. The supervisor and second marker will review the software prototype. The student is expected to demonstrate essential behaviour of the software, and then to explain some part of the code, chosen by the markers;

- Client-based projects will deliver specifications, analyses and reports. The supervisor and second marker will review these specifications and reports to understand their complexity and strengths;
- Research-based projects will deliver a wide range of research outcomes – algorithms, data and analyses of this data, new models and theories etc. These must be reported in a form that makes them available to the supervisor and second marker, for review.

In more detail:

- You have to show your results and the bases on which they were achieved. So, if you have developed software you will usually demonstrate the software in operation, and show some of your code and design documentation; if you have produced specifications, analyses, reports, manuals, data sets and analyses, algorithm descriptions... you will show and explain these.
- prepare a 5-minute basic demonstration of software in operation, or explanation of other outputs and results; the two markers will then ask you to demonstrate or explain this or that aspect of the outputs, to better see their nature and quality
- your second marker will usually need a basic explanation of what your project is about: describe your problem and objectives in a few words, and be ready to answer additional questions so that the rest of your review will be clear
- you must bring all the outputs of your work, to show the markers, and should bring any support documentation which may help you in answering questions thoroughly. So, you will bring software products, source code, specification documents, reports for clients, etc, even if some of this were still in draft form. Support documentation that you must bring also includes documentation of experiment results, questionnaire returns, etc. All the material that will go into the appendices of your project report document must be shown at the review. If you kept a project diary, it is a good idea to have it along in case you want to recall what happened earlier in the project
- be prepared to answer questions about your work: source code, method, data produced. You are the expert on your own project and the markers need the information from you; and the ability to explain your results and method orally is one of the learning outcomes of this module.

9 The Project Report Document

The **Project Report Document** is the principal basis for assessment in your project. As such it demands that you put sufficient effort into its production and delivery. You should start work on the **Project Report Document** as soon as possible, typically by December and submit it to the Department Office, and online as detailed further down, by the set hour on the deadline (cf the calendar on Moodle for your cohort - full time or Professional Pathway).

9.1 Audience and Style

You should assume that your reader has studied computing but not the specific field(s) that your topic fits into. Define any specialist terms, either when they first appear or in a separate glossary. Try to avoid technical jargon and avoid informal abbreviations (e.g. “don’t” for “do not”, “u” for “you”).

Regarding style, write simply, formally and clearly, and do not waste effort trying to achieve a stylish effect. Above all, take care that what you write does convey what you intend it to. Explanations should be detailed enough to be precise without being verbose.

Note that it is easier for you to produce a readable report, and for a reader to appreciate what you did, if you are brief.

Use diagrams to aid communication, but ensure that the reader knows the meanings of symbols in the diagrams (preferably by including a `key' defining their conventions). Use correct spelling and grammar with consistent use of tenses and pronouns. All projects are retrospective reports of what you did and what you found, so write the report in **the past tense**.

Good page layout, accurate and informative headings and cross-referencing will aid the reader to understand your project better.

9.2 Project Report Document. Basic rules

- The main body of a **Project Report Document**, excluding the title page, table of contents, a page with a 100-to-250 word abstract, References and Appendices, must **<u>not exceed 40 pages</u>** (that is, 40 sides of A4 paper; we advise you to produce a double-sided printout to save paper).
- **Formatting instructions.** Use **font Times or Times-Roman 12** and **line spacing 1.5** for all the prose text. You may use different fonts for program code or other special text, but make sure they are large enough to be as readable as the rest. Print the report on A4 paper, with a large enough margin on the side of the binding to allow easy reading. All other margins should be about 2.5 cm wide. It must be possible to open and read every part of your report without undoing any bindings. If the appendices bring the total size of the bound volume to more than 60 sheets, please bind the appendices as a separate volume.
- **Appendices.** Any details that do not fit into the 40-page main body of the report but you expect to be of interest, or that are required to document your work, must be put in appendices. The main body must be a stand-alone document, that is, readers must be able to read through it and find the main information about the project without ever going to the appendices. A common mistake to be avoided is to write the main report as a sequence of references to diagrams, software code and text that are in appendices. Directives about appendices are provided further down in this document.
- **Page count rules** Note that the 40 pages starts from the beginning of your introduction and ends at the end of your conclusions: it excludes the frontispiece, table of contents, any glossary, and the bibliographic reference section.

- **Penalties for excessive length**. A penalty mark will be applied to all **Project Report Documents** that exceed this **Project Report Document** length limit. The penalty will be 1% of the awarded project report mark for each 1 page (or part of a page) that the **Project Report Document** exceeds the maximum page limit. So, for example, if a student is awarded a mark of 60% for the project report, but the report exceeds the 40-page limit by 5 pages, then a penalty of 5% of the awarded mark of 60% will be applied:

  Overall penalty = 5/100 * 60 = 3%

  Therefore the final project report mark will be 60-3% = 57%.

  Furthermore we impose page limits for 3 chapters described in the relevant chapter sections.

- **Page numbering.** All pages must be numbered. Number appendix A's pages A1, A2 etc., and likewise for all other appendices; this helps you to refer to appendix pages whilst still writing the rest of the report. Begin each main section and appendix on a fresh page.

Good page layout, accurate informative headings and cross-referencing can aid the reader to understand your project better.

- **Structure**. Getting the structure of the report right is very important. A useful analogy for you to use is the structure that many of you applied when writing up a scientific experiment at school. These write-ups, which are often written on one side of A4 paper, are divided into introduction, method, results and conclusions. A similar structure can be seen for the project report:

  - Introduction Chapter
  - Method Chapter
  - Results Chapter
  - Conclusions Chapter

- **Usable outputs.** Furthermore, all projects must produce **AT LEAST ONE OUTPUT FOR USE BY A NAMED INDIVIDUAL, GROUP OF INDIVIDUALS OR ORGANISATION**. This output or outputs must be clearly specified in the relevant section of the **Project Report Document**, described below, and included in one or more Appendices of the **Project Report Document**.

The following sections examine the structure of the **Project Report Document** in more detail.

9.3 Introduction

Work from the **Project Definition Document** when writing the Introduction chapter. Describe the underlying problem drivers for the project. These can be elaborated into descriptions of the problem, with any assumptions that you may have used and, were relevant, how you limited the scope of the project, as well as an outline of the work that you did in the project. The Introduction chapter must also present the project objectives, define relevant research hypotheses/questions, and outline the planned work to meet the objectives and answer the research questions.

9.4 Outputs Summary

All projects must produce at least one output for use by a named individual, group of individuals or organisation. The **Project Report Document** must contain a two-page summary of the output(s) produced from the project AFTER THE INTRODUCTION CHAPTER, AND BEFORE THE START OF THE LITERATURE REVIEW. This section of the **Project Report Document** is compulsory, and marks will be awarded when the **Project Report Document** is marked. This summary must be two pages at most (typically a bullet list, one bullet per output) and must contain, for each project output to be produced:

- An overview description;
- what type of output it is – e.g. software code, a system specification, an evaluation report. Be brief and informative, e.g. you might indicate “the product is 10,000 lines of C code (not counting comments), of which 3,000 written by me and the rest re-used”.
- Its recipient and/or end-user;
- How recipients will use the output produced in their work, research or other area of activity;
- A link to each part of the output included in one or more Appendices of the **Project Report Document (e.g., software specifications, software source code, collected data from an experiment);**
- Relevant links to sections of the Results chapter of the **Project Report Document**, identifying how the results described in that chapter contribute to each output.

Not including this section in the **Project Report Document** implies getting zero marks for it.

9.5 Literature Review

The purpose of this chapter is to summarise the reading that you undertook to enable you to do the project work reported in the remainder of the **Project Report Document**. This Chapter must be **at most 3 pages long. It must** summarise the main topics that you read about, and provide full and correct citations to the documents that you have read.

A simple principle for writing this chapter is: consider what are the main sources (books, articles, reference manuals of products you used, documentation of products you had to examine...). For each one, briefly write what information or ideas it contributed that was significant for your work; make sure to list the source itself in the references section, and cite it in your text using the required format for in-text citations. Subdivide the sources by topic (e.g., for a hypothetical design and build project, these topics could be: the application domain of the software; the type of algorithms and architecture required; the language, tools and software libraries used. If there is a client, there may be documents describing their specific domain, the current state of technology in that area of business, etc; for research projects, you would normally have read previous research publications, describing the state of knowledge before your project and helping you to refine the method for running your own project).

In this chapter we are not seeking evidence about how you used this reading – this should be evident from other chapters. However, we are looking for evidence that you have read enough material in the domain and you can summarise the relevance of your readings, and will award marks on this basis.

The length of this chapter must be **3 pages or less**. A penalty mark will be applied if it is longer. The penalty will be 1% mark for each 1 page (or part of a page) in excess of the maximum page limit. The penalty will be applied using the same mechanism for applying length penalties for exceeding the overall **Project Report Document** length.

9.6 Method

The Method chapter provides an <u>objective</u> description of the work undertaken in the project. It should describe the analysis, design, implementation and evaluation processes of your project in detail – be prepared to state the obvious - no detail is too small for the Method chapter and the relevant Appendices. There is sometimes an issue about which version of the project method to present in the Method chapter – a “smoothed out” method that is rewritten at the end of the project, or the “warts-and-all” description of what actually happened during the project. You should give an accurate description of your work, set in the context of a clear general picture of the project plan and general approach.

One consequence of focusing on the details of how you executed the project is that you should avoid reporting more general information about software engineering methods and life cycles: e.g., do not explain the alternative lifecycle models (waterfall etc), but state which one you followed and detail what was specific to your own project. Readers are interested in how *you* applied the standard methods, how you deviated from them and how; and the history of your project, with the main decisions and changes of plans. Do not use the Method chapter as an opportunity to include information that could not fit into the Literature Review chapter. At best this will give you no extra marks, and in the worst case result in marks for the Method chapter being deducted.

One additional but important point about the Method chapter: report what you did, not what you found. Do not report results arising from the application of the method in the Method chapter – that is what the Results chapter is for.

Typically this section will be longer in projects based on fieldwork or data collection than in design-and-build projects. We impose a **maximum page limit** on the length of the Method chapter of **8 pages**. A penalty mark will be applied to all **Project Report Documents** that exceed this chapter length limit. The penalty will be 1% mark for each 1 page, or part of a page, by which the Method chapter exceeds the maximum page limit. The penalty will be applied using the same mechanism for applying length penalties for exceeding the overall **Project Report Document** length.

9.7 Results

The Results chapter presents **ALL OF** the results produced during the project. This can include outputs from analysis, design, prototype, experiment, evaluation and other activities. Again be prepared to state the obvious - no detail is too small for the Results chapter and the relevant Appendices. It is useful to divide the presentation of results into appropriate sections: e.g., for a software development project there may be sections for detailed requirements, software architecture, specific algorithms and coding decisions, verification and evaluation results; for a research project there may be sections about the data collected and their analysis. Be clear in identifying which data you have at the beginning of the project, which ones you collected and which you derived in your further analysis. The results should be traceable to descriptions of each output in the Output Summary chapter, see section 9.4.

We **recommend** that the Results chapter should be at least **20 pages**, as the reported results must be a significant portion of your **Project Report Document**. We will not impose a penalty mark to **Project Report Documents** that do not meet this minimum recommended length limit. However, this length indicates an important focus on the outputs that result from the student's own work in the project.

9.8 Conclusions and Discussion

The Conclusions and Discussion chapter(s) tie-up and discuss all of the project themes. The first thing that you want to do in this chapter is to revisit the project's objectives and research questions and hypotheses. In a successful project you will want to demonstrate that the objectives have been met, the research questions answered, and the research hypotheses accepted or refuted. You will then want to draw more general conclusions from the above. A good project should also discuss the conclusions drawn from the objectives, research questions and research hypotheses. This might happen in 2 parts. Firstly, each conclusion is discussed in its own right. Secondly you should link each conclusion to the literature in the Literature Review chapter, presenting the contributions that your project work has made to our knowledge about the area. The chapter might also discuss implications for future work in the area – perhaps in terms of the next project? Finally, you should also discuss your own progress in the project – the management and control of the project, what you have learned, and what you might do differently if you had your time again.

9.9 Glossary

You may need to define specialist terms, unlikely to be known by your intended audience, in a project glossary appended at the end of the project report.

9.10 Appendices

Any material that would interrupt reading the report should be presented here in the Appendices. These should include **all data used in producing the results of your project and all outputs produced that are not part of the main report, as for instance**:

- Appendix A: **Project Definition Document** (this must always be Appendix A);
- Complete records of each interview; questionnaires and complete questionnaire replies;
- Routine design documentation;
- Source code, line-numbered to aid reference, delivered preferably as a CD rather than a large number of printed Appendices, and with instructions about how to build an executable version;
- Test plans and test results; output listings; displays etc.;
- Any software installation guides and user guides produced;
- An executable version of your software with installation instructions, or a URL if it is a web-based application that can be tested online;
- All the reports produced for a client
- Web pages related to your project.

Rule: complete appendices. Note carefully. All data used in producing the results of your project **must be included** in the Appendices of the **Project Report Document**. Failure to do so will raise concerns about the validity of the results reported: you may be called before the Academic Misconduct Panel to prove that you really did the work described in your report. Therefore, **include** all raw and processed data in the Appendices of the report, and be careful **not to delete or lose** data before the completion of the final year of your degree program. The consequences might be severe.

Non-printed appendices. The appendices must be complete, and normally presented in two copies as the rest of the report, but there are special cases about how to present them:

- electronic appendices: some appendices would be too long if printed (for instance, long source code files, or large data series; an indicative limit for “too long” is 20 pages for one appendix), or easier to read on a computer screen than on paper, or non-visual (e.g., interview records that exist only as audio recordings). If in doubt about these criteria, seek advice from your supervisor. Include these appendices as **electronic files** on one or more CDs or DVDs. The files must be in **formats that can be read with software commonly available on personal computers**, i.e., not require the markers to install any software. Label the disks clearly, and insert them in paper pockets attached to pages in your report or appendices volumes. Make sure there are two copies of each disk, one with each printed copy of the report. Electronic appendices must be also submitted online, together with the appendices you print, typically as a single zip archive. We may set limits on the size of files uploaded for online submission; if these limits give you problems, seek advice from the project tutors
- some appendices are not in electronic format, e.g., hand-written questionnaire answers or interview notes. So, there is a problem with submitting two copies and especially with uploading an electronic copy. If these documents are small they can be scanned and included as electronic appendices. If they are larger, provide only two paper copies. If they are too large (more than 100 pages), an exception can be made to the rule of

submitting two copies. Submit a single copy, with one copy of your report, and indicate in the other copy of the report that this appendix exists in a single copy.

- should you include pieces of source code, design, specification, requirements diagrams, in the main body of the report? This is a frequently asked question.
  This question must be re-worded, putting yourself in the shoes of the readers. The body of the report must be such that readers can understand the main characteristics of the work done, without the help of the appendices. So, among all the material that you *could* include in the 40 pages, you must choose on the basis of its importance. Those diagrams that are most important for understanding the work will usually be in the main body (and possibly repeated in the appendices, so that readers can see the complete design, for instance, in the appendix without jumping back to the main body every now and then). The diagrams that are "most important for understanding the work" are usually not the whole set of diagrams of one type, and they are not all of the same type. For instance, the main body of a report might include:
    - a specific sequence diagram, because it clarifies the way a certain essential protocol works;
    - a specific state machine diagram, because it makes clear some basic idea about how the system evolves;
    - a specific class diagram, because it clarifies some important relationships among data items in the state of the software;
    - and so on, for various diagrams of any kind.
- the same criteria apply to source code. For instance, you might choose to quote in the main body of the report snippets of code that are important for understanding your product, or represent important parts of your original work (e.g., algorithms you invented). Separately, you provide ALL the source code in the appendices, so that readers have access to any part that they wish to read.
- remember that the aspects of the work that most need clear explanations are often those that are so basic for the project that they have become obvious to you: what is the sequence of operations in using a system, what exactly do certain basic data items represent, and so on: you know these facts very well, and may forget that the second marker has not spent seven months on the project as you have.

9.11 Submitting the Project Report Document

You shall submit 2 copies of the **Project Report Document** to the Department Office. Both copies must be identical, with the same content, and be comb-bound (using facilities provided by most commercial printing outlets). Note that we no longer use City's maroon hard cover binders. Your front page must clearly show your name, your project title, the degree you are studying for and the year. We will distribute a prescribed format for this page.

The submitted copies will not be returned to you.

You will also be required to upload an **electronic copy** of your project report, all appendices and program sources you have written and any other documents that may be useful. **Remember that BOTH two paper copies AND an electronic copy must be submitted, by the same deadline.**

The preferred format for the electronic submission is as pdf documents, except for documents that are pure text, in particular source code, which should be submitted as text files, and other special formats like audio files. Rules for bulky appendices have been given above.

You MUST keep the source files (for instance, Word documents, Excel spreadsheets and so on) from which your pdf files were obtained: we may need to examine them. Keep these files until the assessment process has completed and you have received the conclusions of the assessment board by letter.

Late submission will result in the penalties. The Project Report Document will be subject to a reduction of 5% of the final mark awarded for the Document for each day or part day that the report is late – that is submitted after 3pm on the deadline. For example, if a project is submitted 1.5 days after the deadline, then a 10% penalty is applied. Therefore, if the **Project Report Document** receives a mark of 70%, then the penalty applied will be 10/100 * 70 = 7%. As a result the final mark awarded to the **Project Report Document** will be 70-7 = 63%. Note that such penalties are severe, and can have a major impact on the final class of degree that you are awarded, so plan to ensure that you do not miss the submission deadline.

9.11.1 Confidential information

It is possible, especially in a client-based project, that your work uses information that is confidential to your client. In other cases, you may find you have to collect information about individuals (for instance, respondents to a questionnaire; see also the section about Research ethics issues in this document). In all these cases you have a duty to protect the confidentiality of information.

Some of this information may be necessary in your report or appendices. In this case, please notify the course officer before submission. The academics involved in assessing your project will have access to the report, but will know not to cite or divulge the information. **Do make sure that if any volume that you submit contains confidential information to be safeguarded this way is marked as such on its cover.**

It is most convenient for you if any confidential material is in separate, bound appendices: this will make it easier for the School to handle the project report, and you will be able to show the rest of your report, for instance to a possible employer, without violating the confidentiality agreements you made.

10 Good Professional Practice and Ethics

A good project requires hard work, focus on the goals of the project, good thinking, and adherence to professional standards. Here we give essential advice about this last aspect. There are basic rules of sensible and ethical behaviour among professionals: one needs to learn them, and violating some of them is not just a blunder, but a major offence, in the University as well as in professional life.

First you must of course apply the general rules that apply to everyday life: you need to be truthful in what you say and write, deal honestly with clients and anyone else involved, and so on. But in this chapter we call your attention to aspects of good conduct that are very important in professional life and about which you may not have occasion to think much before your project.

Make sure that you read, understand and apply the rules and advice explained in this chapter. If you neglect advice you will lose marks; if you violate the rules, you risk behaving dishonourably, and being punished heavily for it. **It is *your* responsibility to read, understand and follow these rules**; so, if you have any doubt about what they require in your specific circumstances, you must ask for clarifications.

Important note: *if you do not follow the rules and advice given here, your supervisor might notice and warn you about it. This is a fortunate circumstance and you should then take advantage of it to correct all your mistakes or violations,* ***not just*** *those that were noticed.*

10.1 Use other people's results, and *give proper acknowledgment* in the *prescribed way*

All good professionals use the product of other people's work. They read reference books and research articles, they re-use software and take advice from experts. Without this, we would be forever reinventing the wheel. So, in your project you need to read about the methods and technology you need to use, going back to modules studied in the university as well as reading more specialised literature concerning your project topic; seek advice from your supervisor, elder colleagues at a client company, etc. The other side of this rule is that one ***must always acknowledge*** these contributions. Not doing so is discourteous and is also a form of cheating, by letting others believe that you discovered facts, invented designs, wrote material that you have not done. Cheating in one's project carries heavy penalties: it may lead to losing marks, loss of one's honours degree or even expulsion from the University without a degree. More details about misconduct procedures and penalties are in the chapter "The Academic Misconduct Panel, ...".

10.2 Reproducing material from your sources: when and how

You often need to report what someone else said that is pertinent to your topic. The important rules are:

- usually it is best to report in your own words the concepts that are pertinent. In these cases, always make it clear *whose* ideas and results these are. There is a prescribed way of doing so, by an in-text citation pointing to an entry in your reference section. These are explained below, under "References and Referencing Style";
- sometimes one wishes to quote *exactly* what someone else said or wrote. In this case, it is essential to identify clearly **all** the material that is copied word for word. This is done, conventionally, by enclosing it between double quotes. Occasionally you need to reproduce whole paragraphs, in which case one can also typeset them differently from the rest of the text, e.g. indent them with respect to the rest of the text, and must write explicitly that the indented text is taken from your source, giving a reference to the source. Examples are below in the section "References and Referencing Style". Omitting the double quotes and the other indications, even by mistake, is a serious violation of

professional behaviour and thus also of the rules for the project. In some cases it is also a violation of copyright law;

- never copy material while changing words here and there, so that it is more difficult to identify as someone else's. It is a devious practice, it is forbidden, and it is not difficult for the markers to identify. If you really need the other person's own text rather than your own summary, copy it verbatim between double quotes. But keep in mind the next bullet point;
- in general, using a large amount of text from other authors, even if properly acknowledged, is not a good idea. You only have 40 pages for the ideas that *you* have produced using all previous knowledge and many months' worth of your work. Don't waste pages on copies of other people's text.

10.3 Use of copyrighted material

Over the last few years some students have used, on their web sites and in the software produced in their projects, material (e.g. graphics, background music) that they do not have permission from the copyright holder to store or use. If you do this in a web site or in software or other material that you make available to others, you risk being rightly prosecuted for breach of copyright. If you wish to re-use any such material, read carefully whether this use is allowed by the copyright owners, and under what conditions, and respect these conditions.

10.4 Justifying one's decisions, statements and claims

Your project report must document the decisions you took and ideas you developed during your project. However, *it is not enough to say what you thought or felt*. You need to document on which bases you reached a conclusion or took a decision. So, throughout the report, you must support any assertion that you make, either by clear reference to your own work documented in the report (e.g. measurements taken or interview results), or by a reference to indicate its source in the literature. You often also need to explain why this support information (evidence) leads to your conclusion; e.g., why the measurement data imply the judgement that you reached. Take into account that your reasoning may be obvious to you, but not to a reader of your report; pay particular attention to your supervisor's remarks about assertions that you have not sufficiently justified.

Violating this rule will drastically reduce the value of what you write, and thus may cost you many marks.

10.5 References and Referencing Style

Take pains to list all cited references and to cite them clearly in the other sections. Citation of references by surname(s) and date is usually most helpful for the reader and easiest to compile.

The UG project scheme ***strongly recommends*** a standard format for specifying all references and for citing them in the text. This is the **Harvard** style. This style is effective, reader-friendly and easy to use. We especially caution students not to invent their own style: it does not usually work. You can read more about the Harvard style by following the link at:

http://www.lib.monash.edu.au/tutorials/citing/harvard.html

The following information is taken from this web source.

> *Referencing is a standardized method of acknowledging sources of information and ideas that you have used in your assignment in a way that uniquely identifies their source. Direct quotations, facts and figures, as well as ideas and theories, from both published and unpublished works must be referenced. Referencing is necessary to avoid plagiarism, to verify quotations, and to enable readers to follow-up and read more fully the cited author's arguments.*

When citing references within the text of an assignment, use only the name of the author, followed by the year of publication. ***In general, page numbers should be included in all in-text citations, as many schools insist on this practice****:*

The theory was first propounded in 1993 (Hamilton 1994, p. 58)

OR

The theory was first propounded by Hamilton (1994, p. 58).

When referring to two or more texts by different authors, separate them with a semicolon (;):

(Larsen 2000, p. 82; Malinowski 1999, p. 19)

Page numbers are compulsory for quotations:

Larsen (1971, pp. 245-6) noted that "many of the facts in this case are incorrect".

OR

"Many of the facts in this case are incorrect" (Larsen 1971, pp. 245-6).

Note that the text above is indented and we said at the beginning from which source it comes: this is one of the methods for showing the reader which text we have taken directly from this source. The more standard method is to use double quotes, as shown above; but we have not used them to identify the whole block of text we want to identify as taken from our source, because it already contains double quotes, which would cause confusion.

Summary and reminder: all text that you reproduce from your sources must be identified as such, typically between double quotes, so that your reader has no doubt about which text is originally yours and which is not. Any time you use such text, or use ideas from other people, you must insert a proper citation and a corresponding reference entry in your bibliography (or "references") section***. Violating these rules, even by accident, amounts to plagiarism and carries serious penalties.***

Tip: some students have a habit of "writing" their reports by pasting-in pieces of other people's work, and later changing them. ***Don't.*** Even if you mean well, it is as dangerous as going through a store putting merchandise in your own pockets. You may plan to pay at the till, but it is easy to forget, and then you will have stolen the merchandise. ***Only paste-in those sentences that you want to quote word by word, and put them between quotes from the beginning.***

10.6 Re-using software, and how to document re-use

As we indicated, re-using software, that is, including existing software in new products, is an important part of software engineering. To do this ethically, though, you ***must***

- ensure that you are authorised to re-use it (by checking its license), and obey all conditions under which you are granted this permission
- clearly indicate what you re-used and what part of your product it is

So, you **must**:

- in your **methods** chapter, indicate roughly what kinds of products you re-used and how re-use affected your development process.
- In the **results** chapter, list the products you have reused and explain what parts of your own product they form.
- In your **Appendices**, list the libraries used, and flag which parts of the code are your own and which parts are re-used code. Specifically, include at the beginning of your source code a list of all the files, indicating exactly which ones are reused (and the source from which you copied them: give the URL or other way for finding the original code), which ones are completely written by you, and which ones are reused with

changes. For these files where you made changes, you must provide, in addition to your file, a file or printout in which all your changes are clearly visible (that is, one can identify the previous text and your text that replaces it)

- In your conclusions, recall and discuss the role of re-used software in your process and in your product.

It is always a good idea in describing the project's achievements to summarise the size of the product produced (often by more than one measure, e.g., lines of code, number of classes, bytes in executable software, number of web pages, number of database tables, ...) and how it is divided between subsystems. You ***must*** also indicate how much of this is produced by you and how much is re-used.

10.7 Research ethics issues

There is a specific set of concerns and rules about problems that may arise especially when a project (research or not) studies people or data about people. Instructions about these issues are in section 4.8 and the documents referenced there.

11 Project Outcomes: passing or failing

At the end of the project process there are several possible outcomes.

In most cases, you will fulfil all of the requirements of the under-graduate process and pass the project, hopefully with a good mark. Congratulations!

However, failure to attain the pass mark can result in a number of different recommendations being made to the Board of Examiners.

11.1 Failure and referral

If the student has failed to attain the pass mark due to insufficient work, or the work failing to reach the required standard, then the Project Tutors might recommend to the Board of Examiners that the student undertake further work to bring the **Project Report Document** up to the required standard. In such cases, the student will continue to work with the same supervisor. If the work required is minor, the typical deadline for resubmitting the revised version of the **Project Report Document** is near the resit exams of the same year. If the resubmitted **Project Report Document** reaches the required standard, the student will be awarded the basic pass mark of 40% for the undergraduate project module. In some cases, the Project Tutors will decide that more time is needed to bring the work to the required standard, and recommend a submission date for the **Project Report Document** of December of the same year. Note also that this will delay the awarding of your degree, sometimes by up to one year, which in turn can harm your job prospects.

In some circumstances, and in particular if the Academic Misconduct Panel has detected academic misconduct in the project, the Board of Examiners might require you to undertake a new project. The student will need to work with a new supervisor to produce a new timetable for the project. The timetable will require the student to submit **all** deliverables for the new project, from the **Project Definition Document** to the **Project Report Document**. New marks will be awarded for all of these components. The original **Project Report Document** copies submitted will not be returned to you. If the new project work is to the required standard, a basic pass mark of 40% will be awarded for the undergraduate project module. The role of the Academic Misconduct Panel is described in a later section of this document.

If your **second** attempt (resubmission or new project) fails, you will **fail** the IN3007 module. *This typically means not receiving an honours degree.*

11.2 If you failed your individual project at first submission: useful advice

The rules applying have been described above. Here follow some hints and reminders:

- If you were told to improve your current project and resubmit: talk to your supervisor ASAP. Whenever appropriate you will be told a set of improvements that if performed fully would be enough to reach a pass mark. Make sure you understand these fully and achieve them. These required improvements are not a formality. Performing 90% of them will still mean that your resubmission will fail, that is, you will not get an honours degree even if you did well in all other modules.
- Unless the extenuating circumstances panel recommended otherwise, your new mark will be capped at 40%. So, you may think that doing the minimum necessary to achieve 40% is enough. However, many students who plan on this basis fail, because they do not take into account the risk of things not working exactly to plan: they planned just enough work to accomplish a pass mark, ran into some delay or unexpected difficulty, and then it was too late. DO take your supervisor's advice on how high *above* a "bare pass" output you have to aim in order to have a good chance of passing.

- Also consider that the project is about learning. If you failed at first submission, it means that you had failed to learn some of the things required for a good project. Aiming for high quality in your resubmission may be a very good investment of time, because it can give you knowledge that you need and improve your self-confidence, even if you end up with a 40% mark.
- Obey the rules for resubmission. Most importantly, attend a new *final project review* when told; *submit by the deadline*; and submit *all (paper and electronic) copies* required. Some students fail through ignoring these simple rules.
- If you have to do a new project, then you will usually follow the normal project calendar in the next academic year. Make sure you follow the rules that apply in that year, including those concerning how to choose a project and supervisor, the deliverable to produce and the deadlines. Do **not** assume you know all the rules. **Read** the new ones
- If your failure was due to **poor practice or misconduct**, steer very clear of any risk of committing them again, even by mistake. Of course we will pay special attention to detecting any such behaviour; and if you repeat any offence, the penalty will be heavier than the previous time. If your offence was that you used other people's text without acknowledgment, do ***more*** than just removing the passages that we found the first time around. Ask yourself how you wrote your report, and make sure that what you submit as your resubmission is ***all*** your own text, or properly acknowledged quotes.

12 The Academic Misconduct Panel, Plagiarism, and Other Forms of Cheating

The UG project may provide you with a temptation to commit what is known as academic misconduct. Most academic misconduct in projects takes 2 forms – plagiarism and results fabrication. These are unacceptable behaviours and the University takes them very seriously. The punishments for misconduct vary: it may cause a student to receive an Ordinary degree instead of an Honours degree, or even to be expelled from the University without a degree. In any case, misconduct remains in the student's file: if that student graduates and later requires a transcript, the transcript will document that the student cheated. Prospective employers will not like this.

Plagiarism: The UG project can offer temptations to you to plagiarise or make mistakes in the presentation of other people's work so that it, or some of it, will appear to be your own work. However, you must beware of these temptations. If you plagiarise, you are likely to be caught using the mechanisms that we have in place to catch plagiarism. Even for small offences, the punishment will both damage your final degree classification (you may receive a lower class of honours degree or an Ordinary instead of an Honours degree) and delay when your degree is awarded. Every year we catch students – 20 or so students out of 180 from the 2004-5 academic year were pulled up for academic misconduct of some form – and some of the punishments resulting from the Academic Misconduct Panel are extreme. Please do not be one of these students in 2010-11.

Results fabrication: Students under pressure – of time or the need for better results – have sometimes made up results. This can take many forms. Students invent responses to questionnaires, generate fictional interviewees who provide them with data, run focus groups with friends and family who are not who they are claimed to be, present software code that appears to function but does not, and so on. Making up results like this is not clever. The members of the academic staff have seen it all before and know what to look for. We catch most such cases and bring them before the Academic Misconduct Panel. If you were tempted, remember that fraudulent behaviour may cost you your project credits (the equivalent of three modules) or even your degree. If you are running late, **talk** with your supervisor for advice. Honest failures may cost you marks, but dishonesty can cost a student's degree and ruin a career before it can start.

12.1 The Academic Misconduct Panel (AMP)

The School of Informatics has set up an AMP specifically to handle academic misconduct detected in UG projects. It is a formal body comprising the 2 main Project Tutors, one other member of the academic staff, and a senior member of the UG office in the Department of Computing.

The University Assessment Regulations define 'Academic Misconduct' as follows:

> "Academic misconduct is any action that produces an improper advantage for the candidate in relation to his or her assessment or deliberately and unnecessarily disadvantages other candidates. It includes, but is not limited to, such offences as plagiarism, impersonation, collusion and disruption."

The Assessment Regulations provide for the student to have the opportunity to present his or her case to the Panel. Presentation may take the form of:

- A written submission from the student (with comments invited from the Personal Tutor) or;
- A meeting with the Panel with the Personal Tutor present.

The AMP will produce a written record in the form of minutes and recommendations to the Board of Examiners.

12.2 Allegations of Misconduct Relating to Plagiarism

The ever-increasing use of the internet offers both opportunities to students to copy and/or plagiarise other people's work that is available on the internet and software tools that enable academic staff to discover plagiarised work. We have recently added new software tools to our armoury for checking suspicious segments of the **Project Report Document** or other outputs from the project. All supervisors are under instruction to use these tools to investigate possible plagiarism throughout the project. The AMP will consider cases of suspected plagiarism.

The AMP will not take account of any claimed extenuating circumstances in plagiarism cases.

The AMP will reach one of the following decisions for each case brought before it.

12.2.1 No Academic Misconduct Has Taken Place

The AMP may decide that no misconduct has taken place, in which case the student is informed in writing, and the allegation will not be recorded on the student's file.

12.2.2 Poor Practice Has Taken Place

The AMP may decide that no misconduct has taken place but there is evidence of poor practice that should be taken into account by the marker. In this case an administrative check should be made after the panel that the marks awarded did take account of the poor practice (i.e. the marker should be asked to confirm that the final awarded mark took account of the poor practice; if not, the final awarded mark should be reduced accordingly).

Typical examples of poor practice include minor cases of poor referencing and inadequate reporting of the work undertaken:

The penalty for incorrect referencing that does not amount to misconduct is a deduction of 5% from the mark of the **Project Definition, Project Progress and Project Report Documents** for **each separate instance** of the incorrect referencing identified in the report. This deduction might well impact on the final class of degree that is awarded to you. Furthermore, the Project Tutors may require the student to fix the referencing and citations in the submitted document. If this occurs with the **Project Report Document** this delay might well cause a delay in the awarding of the final degree, with possible implications for your graduate employment. Furthermore, if incorrect referencing is systematic, it will raise suspicions of major plagiarism, described in the next section.

Penalties for failing to report all of the work and results, or misrepresentation of the work and results reported can be more severe depending on the nature of the offence, and can lead to a mark of 0 being awarded for the relevant section of the **Project Report Document**.

The student will be informed in writing. The allegation of misconduct will not be recorded on file. However, poor practice should be avoided and future occurrences might lead to consequent penalties.

12.2.3 Academic Misconduct Has Taken Place

The AMP may decide that misconduct has taken place. The usual penalty to be recommended to the Assessment Board will be a mark of zero for the submitted work. In the case of a second finding of academic misconduct for a student, the Panel may refer the case to the Academic Registrar as specified in the Assessment Regulations. The student will be informed in writing that the Panel has decided that misconduct has taken place, with a note of the penalty to be recommended to the Assessment Board. This is a recommendation only and final decisions will be made by the Assessment Board. There is no right of appeal against this judgment by the AMP.

Academic misconduct includes major plagiarism and fabrication of results and of other outputs.

Examples of major plagiarism include:

- Consistent failure to cite the work of others in the Method, Results and Conclusion and Discussion chapters of the **Project Report Document**;
- Presenting the results or work of others as your own, either deliberately or unintentionally;
- Unacknowledged reuse of work from another source in the completion of your project**.**

Typical examples of work fabrication include:

- Presenting the results of data gathering from questionnaires, interviews, evaluations, etc in order to deceive the Project Tutors, Supervisor and Second Marker as to the source of the data;
- Reporting a software solution as fully functional when in reality the functionality of the software is defective or incomplete, in order to deceive the Project Tutors, Supervisor and Second Marker.

Incidents of major plagiarism will be processed following the University regulations on plagiarism. This may well cost the whole project mark, that is, make it impossible to get a honours degree.

Incidents of work fabrication will be processed in a similar way.

As passing the undergraduate project module is a pre-requisite for passing the final year of your degree course, the implications are stark. Take heed.

The University provides further information on plagiarism.

Appendix A: Client information sheet, to be Submitted with Project Definition Document at Start of Project

If you have a client-based project, with your **Project Definition Document** you must submit a description of your client and your planned way of working with your client. This is an **extra** page at the end of each copy of your PDD.

Not attaching this page implies receiving **0 marks** for your PDD. The School will send this page to the client and ask them to confirm that the information is correct and that they agree to be your client in this project. This helps us to be sure that you have a committed client, and protects you by having the company confirm their agreement with you in writing, to avoid misunderstandings. This is why we need this information **repeated** clearly in this page, although it is also required elsewhere in your PDD.

The information required is:

- complete name of the company and contact information (address, telephone and fax number, Email address). As a general rule, we want company Email addresses (not hotmail, for instance), and landline numbers for the company's official premises.
- name[s] and contact information for your contact person[s], if different.
- any form of relationship between you and the company (e.g. if the owner or your contact person is a relative, or you are employed by the company)
- topic of the project: what output it will produce. One or two lines are usually enough
- what assistance the company will give you: e.g., interviews with staff; supervision by a member of staff; software to re-use; use of company premises or computers. Be specific. You need to have a clear common understanding with your client, or you risk disputes later which may endanger your project;
- confirmation that the company will provide a letter at the end of the project to confirm what input they provided and to indicate their assessment of the project outputs. In other words, you must have informed the client of this requirement and they must have accepted it.

Remember to attach a copy of this documentation to each copy of your **Project Definition Document**.

If any of this information changes during the year, you need to inform your project supervisor and your course officer.

Appendix B: The Supervisor Charter

Academic supervisors play a very important role in student projects. For this reason we have produced a simple charter that specifies the minimum level of involvement that you should expect from your supervisor during the project. If you believe that your supervisor is not providing this minimum level of supervision you should contact the relevant Project Tutor.

The following specifies the minimum level of support that you should receive from your project supervisor:

- If your project title has been provided by your project supervisor, your project supervisor should indicate background reading material on the project or its application;
- Your project supervisor should read and provide feedback on at least one complete draft of your **Project Definition Document** prior to its submission, if given to the supervisor at least 72 hours prior to the submission deadline;
- Your project supervisor should read and provide feedback on at least one complete draft of your **Project Report Document** prior to its submission deadline, if the draft is submitted to the supervisor by the required deadline;
- Your project supervisor should be available to meet with you once in every 3 weeks, during teaching terms, for the duration of the project.